Skills Summary

Solving One-Step Equations with Rational Coefficients

Use this reference while completing your worksheet or when studying.

One step in, one step out. An equation $cx = k$ says "x was multiplied by c, and the result is k." Undo that one multiplication on both sides.

- c is a **fraction** → multiply both sides by its **reciprocal** (flip it). The reciprocal of $\frac{a}{b}$ is $\frac{b}{a}$, and a negative coefficient keeps its sign when flipped.
- c is a **decimal** → divide both sides by c.
- If c is less than 1, the solution is *bigger* than k; if c is bigger than 1, the solution is *smaller* than k. Predict this first — it catches most mistakes before they happen.

1. Unit-fraction coefficient: multiply by the denominator

$\dfrac{x}{b} = k$ means the bar x was cut into b equal parts and one part is k. The reciprocal of $\frac{1}{b}$ is just b, so multiply both sides by b.

Example: Solve $\dfrac{x}{5} = 8$.
Step 1. One fifth of the bar is 8, so the whole bar is five of those parts — multiply both sides by 5 rather than guessing.
Step 2. $5 \cdot \dfrac{x}{5} = 5 \cdot 8$, and the 5s on the left cancel.

$$x = \mathbf{40}$$

Try it: Solve $\dfrac{x}{6} = 7$.

check: **42**

2. Decimal coefficient: divide both sides

$0.6x = k$ was built by multiplying, so divide. If the division is awkward, multiply the numerator and denominator by 10 or 100 to clear the decimal: $\dfrac{9}{0.6} = \dfrac{90}{6}$.

Example: Solve $0.6x = 9$.
Step 1. The coefficient is less than 1, so predict a solution larger than 9, then divide to find it.
Step 2. $x = \dfrac{9}{0.6} = \dfrac{90}{6}$, which is 15 — larger than 9, as predicted.

$$x = \mathbf{15}$$

Try it: Solve $0.25x = 7$.

check: **28**

3. Non-unit fraction: multiply by the reciprocal

$\dfrac{a}{b}x = k$ means a of the b equal parts are worth k. Multiplying by $\dfrac{b}{a}$ turns the coefficient into 1 in a single step — it is the same as dividing by a and then multiplying by b, done at once.

Example: Solve $\dfrac{3}{4}x = 15$.

Step 1. Three of four parts are 15, so flip the coefficient and multiply both sides by the reciprocal.

Step 2. $\dfrac{4}{3} \cdot \dfrac{3}{4}x = \dfrac{4}{3} \cdot 15 = \dfrac{60}{3}$.

$$x = \mathbf{20}$$

Try it: Solve $\dfrac{2}{5}x = 14$.

check: **35**

4. Check by substituting, not by re-solving

To check a solution, put the value back into the *original* equation and evaluate the left side. If it equals the right side, the value is a solution. Do not use the reciprocal here — that is the solving move, not the checking move.

Example: Is $x = 20$ a solution of $\dfrac{3}{4}x = 15$?

Step 1. Checking means substituting into the equation exactly as written, so use the coefficient $\frac{3}{4}$, not its reciprocal.

Step 2. $\dfrac{3}{4} \cdot 20 = \dfrac{60}{4}$, which is 15 — matching the right side, so 20 is a solution.

$$\frac{3}{4} \cdot 20 = \mathbf{15}$$

Try it: Check $x = 35$ in $\dfrac{2}{5}x = 14$ by substituting.

check: **14**

(!) Watch out: multiplying by the coefficient instead of its reciprocal is the single most common error — $\frac{2}{3}x = 18$ does *not* give $x = \frac{2}{3} \cdot 18$. And keep the sign with the reciprocal: the undo of $-\frac{3}{5}$ is $-\frac{5}{3}$.